



UNIVERSITY OF TECHNOLOGY SYDNEY

PRICING REPORT GROUP 5

Pricing

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1 Task 1

We have the following assumptions:

- **Deterministic Interest Rates:** The continuously compounded short rate $r(t)$ is deterministic and time-dependent
- **Continuous Dividend Rate:** The S&P 500 index pays dividends at a continuous dividend rate $q(t)$, which is also deterministic and time-dependent.

We can also define mid prices as:

- **Call Mid Price:** $\frac{Bid_{call} + Ask_{call}}{2}$
- **Put Mid Price:** $\frac{Bid_{put} + Ask_{put}}{2}$

In order to determine the values of $B(0, T)$ and $D(0, T)$ that best satisfy the put-call parity, we minimized the sum of squared deviations (SSD) from the parity condition. Specifically, for each maturity, the objective was to minimize:

$$SSD = \sum ((C(K, T) - P(K, T)) - (D(0, T)S(0) - B(0, T)K))^2 \quad (1)$$

In python we implement it using the minimize function from the SciPy library. For each maturity:

- Initial values for $B(0, T)$ and $D(0, T)$ were set to 0.9.
- Constraints were added to ensure the sum of the deviations from the parity condition was close to zero.
- The squared deviation function was minimized for each unique maturity in the dataset.

The resulting term structure, provides the optimal values of $B(0, T)$ and $D(0, T)$ for each maturity T in the dataset. The result can be found in the python file.

2 Task 2

The implied volatility for each maturity T_i is determined by calibrating the Black-Scholes model so that the theoretical price equals the market price for options at that maturity. In this context, we assume:

- A deterministic risk-free rate function.
- A deterministic continuous dividend yield.
- Piecewise constant volatilities on intervals defined by option maturities.

The forward price of the index for each maturity, given by $S(T)$, is computed using the zero-coupon bond price $B(0, T)$ and the dividend discount factor $D(0, T)$:

$$S(T) = S_0 \frac{D(0, T)}{B(0, T)} \quad (2)$$

where S_0 is the index level on the valuation date.

Using the forward prices, we select strikes closest to these forward prices as the basis for calculating the implied volatilities. To compute the implied term structure of volatility:

- For each maturity, choose the strike price closest to the forward price.
- Use Black-Scholes pricing to match observed bid and ask prices with theoretical prices derived from implied volatilities.
- Determine $\sigma_{bid}(t)$ and $\sigma_{ask}(t)$ such that the resulting piecewise constant volatility functions satisfy:

$$C = S(0)D(0, T)N(h1) - KB(0, T)N(h2) \quad (3)$$

Now we are going to prove (3).

Proof.

$$S(T) = S_0 \exp \left(\int_0^T \left(r(t) - q(t) - \frac{1}{2} \sigma(t)^2 \right) dt + \int_0^T \sigma(t) dW(t) \right)$$

However, since $\sigma(t)$ is a piecewise function, we can express it as:

$$\int_0^T \sigma(t) dt = \sum_{i=1}^n \int_{T_{i-1}}^{T_i} \sigma_i(t) dt$$

Moreover, $\sigma_i(t)$ is constant on the intervals $[T_{i-1}, T_i]$, so we have:

$$\sum_{i=1}^n \int_{T_{i-1}}^{T_i} \sigma_i(t) dt = \sum_{i=1}^n \sigma_i(T_i - T_{i-1})$$

And we can express the discount factors as:

$$\exp \left(- \int_0^T r(t) dt \right) = B(0, T)$$

$$\exp\left(-\int_0^T q(t)dt\right) = D(0, T)$$

This leads us to:

$$S(T) = S(0) \frac{D(0, T)}{B(0, T)} \exp\left(-\frac{1}{2} \sum_{i=1}^n \sigma_i^2 (T_i - T_{i-1}) + \sum_{i=1}^n \sigma_i \int_{T_{i-1}}^{T_i} dW(t)\right)$$

Now, the payoff for a European call option is given by:

$$C = \mathbb{E}_\beta \left[\max(S(T) - K, 0) \exp\left(-\int_0^T r(t)dt\right) \right]$$

This can be rewritten as:

$$C = \exp\left(-\int_0^T r(t)dt\right) \mathbb{E}_\beta [S(T) \mathbb{I}(S(T) > K)] - \exp\left(-\int_0^T r(t)dt\right) K \mathbb{E}_\beta [\mathbb{I}(S(T) > K)]$$

Which simplifies to:

$$C = \exp\left(-\int_0^T r(t)dt\right) \mathbb{E}_\beta [S(T) \mathbb{I}(S(T) > K)] - \exp\left(-\int_0^T r(t)dt\right) K \mathbb{P}_\beta(S(T) > K)$$

Now, using the properties of the normal distribution, we have:

$$\mathbb{P}_\beta(S(T) > K) = \Phi\left(\frac{\ln\left(\frac{S(0)}{K}\right) + \ln\left(\frac{D(0, T)}{B(0, T)}\right) - \frac{1}{2} \sum_{i=1}^n \sigma_i^2 (T_i - T_{i-1})}{\sqrt{\sum_{i=1}^n \sigma_i^2 (T_i - T_{i-1})}}\right)$$

Next, we observe that:

$$\frac{S(T)}{S(0)} \cdot \frac{D(0, T)}{B(0, T)} = \exp\left(-\frac{1}{2} \sum_{i=1}^n \sigma_i^2 (T_i - T_{i-1}) + \sum_{i=1}^n \sigma_i \int_{T_{i-1}}^{T_i} dW(t)\right)$$

This represents an exponential martingale. By applying Girsanov's theorem, we can change the measure from $\mathbb{P}$ to $\mathbb{Q}$

We then have:

$$W_Q(T_i) - W_Q(T_{i-1}) = W(T_i) - W(T_{i-1}) - \sigma_i(T_i - T_{i-1})$$

Consequently, $S(T)$ becomes:

$$S(T) = S(0) \frac{D(0, T)}{B(0, T)} \exp\left(+\frac{1}{2} \sum_{i=1}^n \sigma_i^2 (T_i - T_{i-1}) + \sum_{i=1}^n \sigma_i (W_Q(T_i) - W_Q(T_{i-1}))\right)$$

Thus, we find:

$$\mathbb{E}_\beta [S(T)\mathbb{I}(S(T) > K)] = \frac{S(0)D(0, T)}{B(0, T)} \mathbb{E}_Q [\mathbb{I}(S(T) > K)] = \frac{S(0)D(0, T)}{B(0, T)} \mathbb{P}_Q(S(T) > K)$$

And we express this as:

$$= \frac{S(0)D(0, T)}{B(0, T)} \Phi \left(\frac{\ln \left(\frac{S(0)}{K} \right) + \ln \left(\frac{D(0, T)}{B(0, T)} \right) + \frac{1}{2} \sum_{i=1}^n \sigma_i^2 (T_i - T_{i-1})}{\sqrt{\sum_{i=1}^n \sigma_i^2 (T_i - T_{i-1})}} \right)$$

Finally, we arrive at the expression for the call option price:

$$C = S(0)D(0, T)N(h_1) - KB(0, T)N(h_2)$$

Where $h_{1,2}$ are given by:

$$h_{1,2} = \frac{\ln \left(\frac{S(0)}{K} \right) + \ln \left(\frac{D(0, T)}{B(0, T)} \right) \pm \frac{1}{2} \sum_{i=1}^n \sigma_i^2 (T_i - T_{i-1})}{\sqrt{\sum_{i=1}^n \sigma_i^2 (T_i - T_{i-1})}}$$

□

So in python we used the fsolve function from SciPy to solve for $\sigma_{bid}(t)$ and $\sigma_{ask}(t)$, ensuring that the resulting Black-Scholes prices matched the observed bid and ask prices, respectively. The result could be find in the python file.

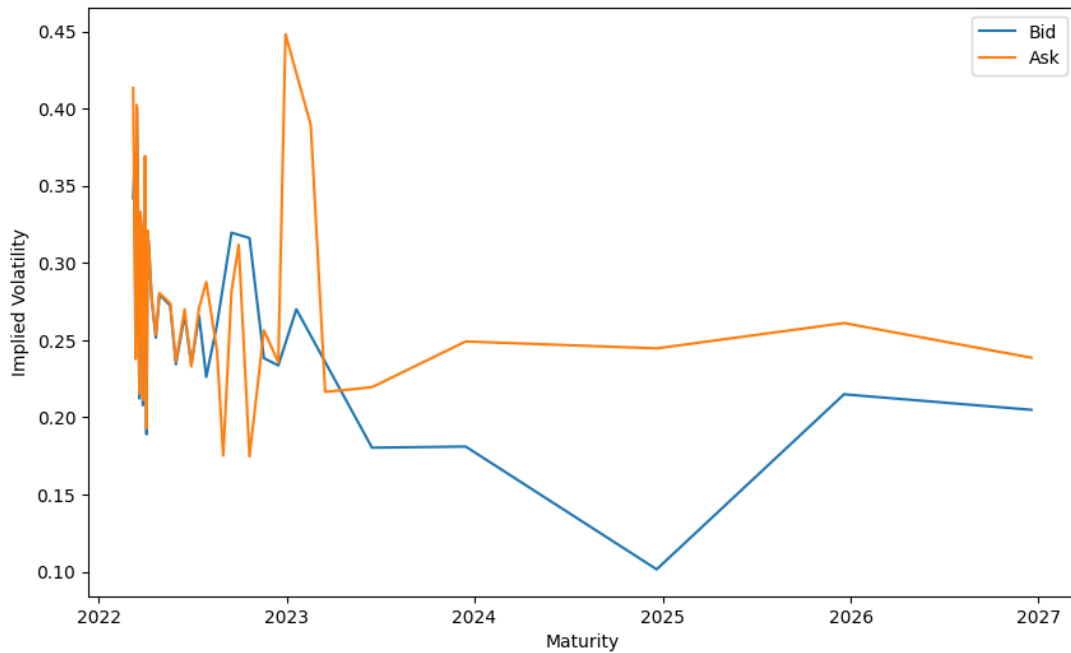


Figure 1: Bid and Ask implied volatility

3 Task 3

We want to derive the formula for the time 0 price of the following payoff:

$$\max(S(0)e^{gT}, S(0)^\alpha S(T)^{1-\alpha})$$

Proof. So we can write:

$$\begin{aligned} V &= B(0, T) \mathbb{E}_\beta[\max(S(0)e^{gT}, S(0)^\alpha S(T)^{1-\alpha})] \\ &= B(0, T) \mathbb{E}_\beta[S(0)e^{gT} \mathbb{I}(S(T) \leq S(0)e^{\frac{gT}{1-\alpha}})] + B(0, T) \mathbb{E}_\beta[S(0)S(T)^{1-\alpha} \mathbb{I}(S(T) > S(0)e^{\frac{gT}{1-\alpha}})] \end{aligned}$$

First term:

$$\begin{aligned} B(0, T) \mathbb{E}_\beta[S(0)e^{gT} \mathbb{I}(S(T) \leq S(0)e^{\frac{gT}{1-\alpha}})] &= B(0, T) S(0) e^{gT} \mathbb{P}(S(T) \leq S(0)e^{\frac{gT}{1-\alpha}}) \\ &= B(0, T) S(0) e^{gT} N \left(\frac{(\frac{gT}{1-\alpha}) - \ln(\frac{D(0, T)}{B(0, T)}) + \frac{1}{2} \int_0^T \sigma^2(t) dt}{\sqrt{\int_0^T \sigma^2(t) dt}} \right) \\ &= B(0, T) S(0) e^{gT} N(h_1) \end{aligned}$$

Then we have:

$$\begin{aligned} cS(T)^{1-\alpha} &= c(S(0) \frac{D(0, T)}{B(0, T)})^{1-\alpha} \exp \left((1-\alpha) \left(-\frac{1}{2} \int_0^T \sigma(t)^2 dt + \int_0^T \sigma(t) dW(t) \right) \right) \\ &= c(S(0) \frac{D(0, T)}{B(0, T)})^{1-\alpha} \exp \left(-\frac{1}{2} \int_0^T (1-\alpha) \sigma(t)^2 dt + \int_0^T (1-\alpha) \sigma(t) dW(t) \right) \end{aligned}$$

is an exponential Martingale iif:

$$\begin{aligned} c &= \left(\frac{B(0, T)}{D(0, T)} \right)^{1-\alpha} \exp \left(\frac{1}{2} \int_0^T (1-\alpha) \sigma(t)^2 dt - ((1-\alpha) \sigma(t))^2 \right) \\ &= \left(\frac{B(0, T)}{D(0, T)} \right)^{1-\alpha} \exp \left(\frac{1}{2} \int_0^T \alpha (1-\alpha) \sigma(t)^2 dt \right) \end{aligned}$$

So the second Term is:

$$\begin{aligned} B(0, T) \mathbb{E}_\beta[S(0)S(T)^{1-\alpha} \mathbb{I}(S(T) > S(0)e^{\frac{gT}{1-\alpha}})] &= \\ B(0, T) \mathbb{E}[S(0) \left(\frac{D(0, T)}{B(0, T)} \right)^{1-\alpha} \exp \left(-\frac{1}{2} \int_0^T \alpha (1-\alpha) \sigma(t)^2 dt \right) \mathbb{I}(S(T) > S(0)e^{\frac{gT}{1-\alpha}})] &= \\ S(0) D(0, T)^{1-\alpha} B(0, T)^\alpha e^{\frac{\alpha^2 - \alpha}{2} \int_0^T \sigma^2(t) dt} \mathbb{E}[\mathbb{I}(S(T) > S(0)e^{\frac{gT}{1-\alpha}})] \end{aligned}$$

Applying Girsanov Theorem:

$$dW(t) = d\bar{W}(t) + (1-\alpha)\sigma(t)dt$$

Yielding:

$$S(T) = S(0) \frac{D(0, T)}{B(0, T)} \exp \left(\left(\frac{1}{2} - \alpha \right) \int_0^T \sigma(t)^2 dt + \int_0^T \sigma(t) d\bar{W}(t) \right)$$

So finally we have the following formula:

$$V = S(0)B(0, T)e^{gT}N(h_1) + S(0)D(0, T)^{1-\alpha}B(0, T)^\alpha e^{\frac{\alpha^2 - \alpha}{2} \int_0^T \sigma^2(t)dt} N(h_2) \quad (4)$$

Where:

$$h_1 = \frac{\left(\frac{gT}{1-\alpha}\right) - \ln\left(\frac{D(0,T)}{B(0,T)}\right) + \frac{1}{2} \int_0^T \sigma^2(t)dt}{\sqrt{\int_0^T \sigma^2(t)dt}}$$

$$h_2 = \frac{-\left(\frac{gT}{1-\alpha}\right) + \ln\left(\frac{D(0,T)}{B(0,T)}\right) + \left(\frac{1}{2} - \alpha\right) \int_0^T \sigma^2(t)dt}{\sqrt{\int_0^T \sigma^2(t)dt}}$$

□

4 Task 4

We use the formula for the price at the time 0 of the payoff which we found on Task 3. Then we use `fsolve` function from SciPy to find the g associated to a specific α . Here is a table that sum up the results we found:

α	0.25	0.5	0.75
g	-0.0218	0.0047	0.0196

Table 1: Values of g for various α

5 Task 5

To solve Task 5, we used the `minimize` function from SciPy to find the weights of the portfolio that minimize its value. However, one of the requirements was to ensure a positive payoff regardless of the outcome. To achieve this, we extracted 10,000 points across the interval $[0, 2 \times \text{Last Strike}]$ and computed the difference between the replicating option payoff and the actual option payoff.

Using a non-negativity constraint did not yield optimal results; therefore, we shifted our method and are now verifying that the sum of squared differences of negative values is close to zero. We applied a loop for this optimization, using the previous results. Moreover, we noticed that some bid prices for calls are equal to zero, and we did not want to include these in the optimization. Instead of setting this as a bound in the optimization, we enforced a weight of zero if the found weight was negative, which we found to be the most effective method.

Finally, since the optimization was performed on only 10,000 points for efficiency purposes, the replicating payoff curve could be below the option payoff at some local points. To address this, we used the `differential_evolution` function from SciPy to find the smallest difference, and if this difference was negative, we added an equivalent quantity of Zero-Coupon Bonds (ZCB) to the portfolio to adjust the payoff.

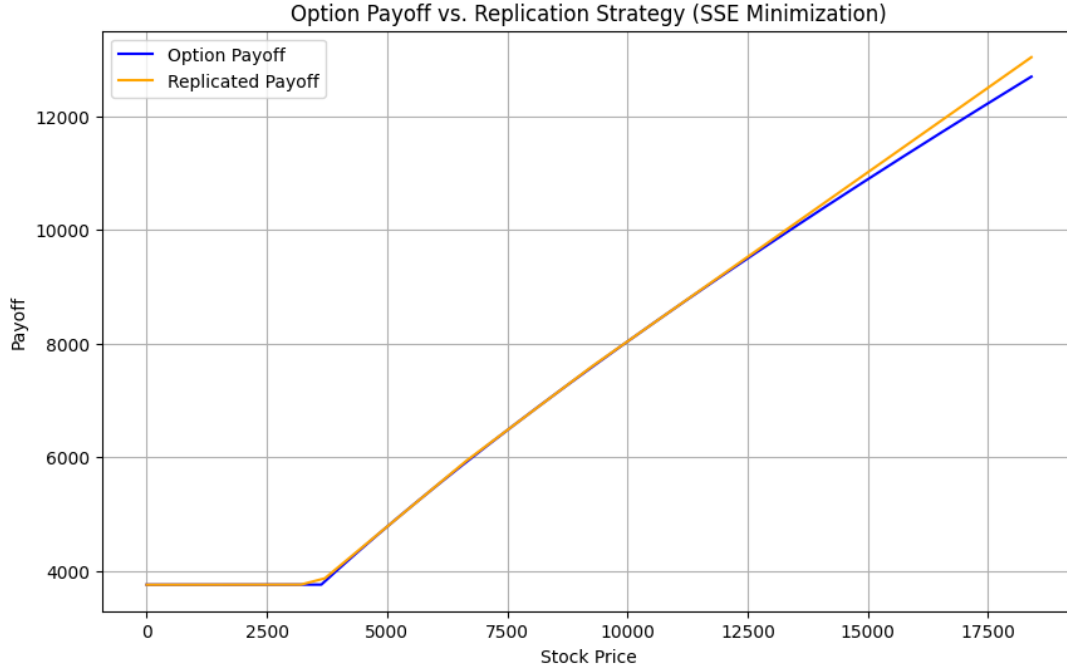


Figure 2: $\alpha = 0.25$, replication cost = 4259.896

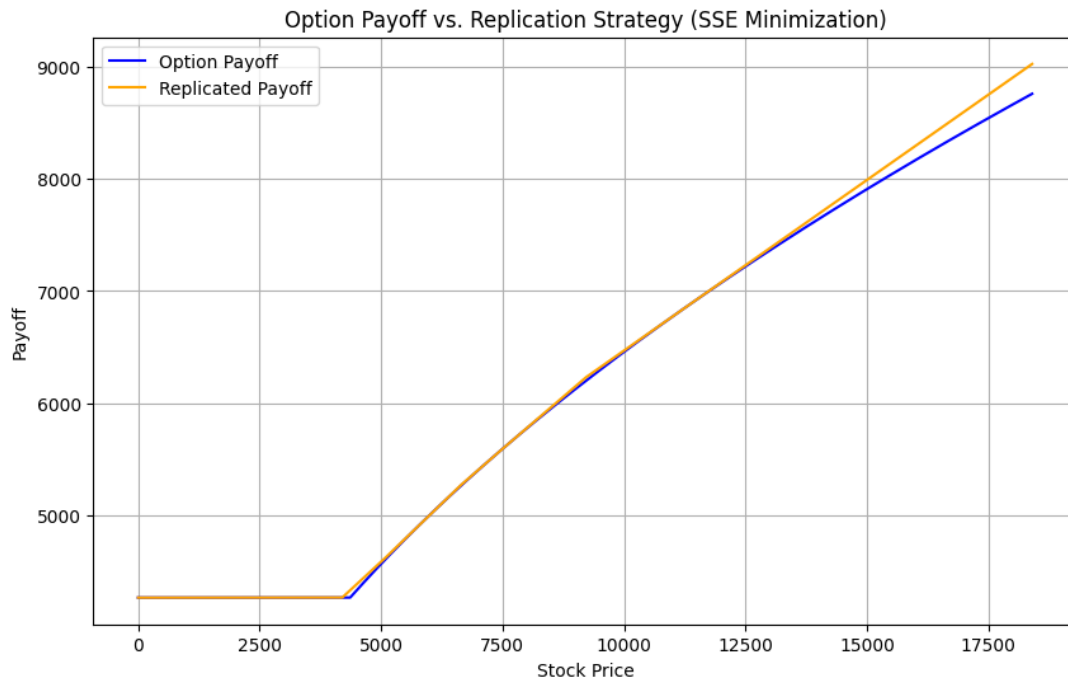


Figure 3: $\alpha = 0.5$, replication cost = 4191.914

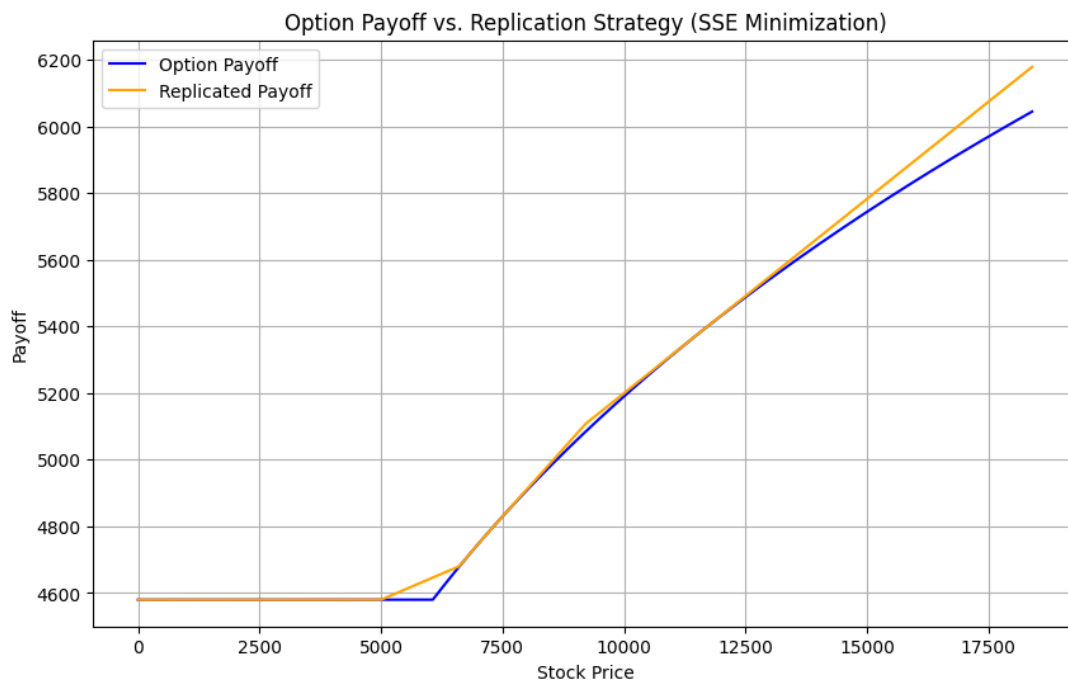


Figure 4: $\alpha = 0.75$, replication cost = 4170.323

The Twelve first values are call weights ordered by strike values. The next twelve values are the put weights and finally the last weight is the number of ZCB bought.

Table 2: Weights

Index	$\alpha = 0.25$	$\alpha = 0.5$	$\alpha = 0.75$
1	0.010878329569588398	-5.164107613080563e-09	0.03614263221536021
2	0.21626737813029223	1.1010127780962557e-06	1.8231593833061715e-08
3	0.48710387380022513	-1.50866932955117e-07	3.924248667810527e-08
4	4.7848112449503846e-06	1.3427141324356403e-09	-1.3485501872189527e-08
5	-6.889430809041603e-07	0.3976823422587101	8.99838419987732e-08
6	-6.330899408776787e-06	0.020191388436860445	0.06226756640130639
7	-0.06659131616924584	-0.038138345694126434	0.10704653414033716
8	6.687640524903606e-07	-0.007487872020088691	-0.006627995753276872
9	0.0	0.0	0.0
10	7.59181347712205e-08	0.0	0.0
11	0.0	0.0	0.0
12	-0.05242222113228319	-0.06789222534354296	-0.08228624574456851
13	-0.010873313356082983	-3.0541977346561207e-07	-0.036142469830553706
14	2.3031034548499567e-07	-1.3475434933328107e-08	-1.4490791110309774e-08
15	1.4467878708202926e-06	-8.06964356616352e-08	-1.223193721784853e-07
16	1.9607652115514168e-07	2.373045556928459e-10	-3.486043071112592e-08
17	6.419711340458212e-09	0.00014136198322173678	-7.590032103430805e-09
18	-1.742571710883837e-07	1.3834748007973877e-06	3.7126714031193955e-09
19	2.3108010622947986e-08	-9.418398629800034e-07	3.5672557671230157e-06
20	5.075597650818985e-06	-7.514405573761331e-07	-4.958314378089213e-09
21	1.534765251972323e-07	-2.438512299617225e-07	6.154405487924885e-09
22	1.3766375341925756e-06	-4.710365538630471e-06	-2.2788043207902697e-08
23	-5.050500218026451e-09	-9.565008425221546e-08	2.5856256239612927e-08
24	0.010866566374123733	-0.000135344327869433	0.03613919850387188
25	3684.4798952802203	4266.848139028673	4334.016118125524

6 Task 6

In Task 4, we derived the "fair" guarantee level g for different values of α . These results show that as α increases, g also increases. This trend can be explained by the structure of the contract: as α (the client's share in any realized ex-dividend return above g) increases, the bank can afford to set a higher guarantee level g without incurring excessive risk. A higher α implies that the client has a greater share in returns above the guaranteed level, so the bank's liability is reduced if the returns exceed g . Consequently, with higher α , the investment bank can set a higher g , making the contract more attractive to the client while managing its own risk exposure.

In Task 5, we calculated the replication cost of the contract for the same values of α . So we see that the replication cost decreases as α increases. This is because, with a higher α , the bank bears less risk if the returns exceed g , as the client takes a larger share of the upside. As a result, the bank requires a smaller cost to replicate the guaranteed payoff.

In summary:

- Higher α allows the bank to set a higher g while lowering the replication cost.
- Lower α requires a lower g but increases the bank's replication cost, as it retains more risk in the contract.